SHAZIN

Final-Year Computer Engineering Student | Embedded Systems, Robotics & Software

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PROFILE

Final-year Computer Engineering student with a 3.63/4.00 CGPA and two Vice-Chancellor's List awards. Builds across embedded firmware, autonomous systems, discrete-event simulation and full-stack software, with a record of independent learning and open-source collaboration. Research interests include low-power sensor systems, heterogeneous robot coordination, simulation-based optimisation and practical Linux systems.

EDUCATION & ACADEMIC HONOURS

Asia Pacific University of Technology & Innovation (APU) — **2022 - Expected 2026**

Bachelor of Engineering in Computer Engineering | Final semester

- Current CGPA: 3.63/4.00 through Year 4, Semester 1.
- Vice-Chancellor's List for Excellent Academic Achievement: Academic Year 2023/2024 and Academic Year 2024/2025.

INDUSTRY EXPERIENCE

Averis - Engineering Intern (Embedded Systems & IoT) — **Jun - Oct 2025**

Low-power sensor networks, satellite connectivity, prototyping and technical documentation

- Developed C++ firmware for custom STM32U0 ground nodes that read water-level sensors over Modbus/RS-485, pack telemetry and transmit it through an Iridium Short Burst Data satellite modem.
- Engineered low-power and fault-tolerant behaviour including RTC shutdown/deep sleep, battery monitoring, automatic sensor detection, FRAM/EEPROM fallback storage and queued retransmission after connectivity failures.
- Validated ground nodes in field deployments; documented handover material and guided incoming interns. Also standardised FDM printer calibration and OrcaSlicer profiles and prototyped node enclosures.

ENGINEERING PROJECTS

Agricultural Robot Fleet Discrete-Event Simulator - Final Year Project — **2026 - Present**

Simulation-based optimisation of heterogeneous oil-palm harvesting fleets

- Developing a simulation to determine efficient fleet composition and charging-station capacity for Langur harvesters, Howler collectors and Mandril transporters operating as a dependency-linked pipeline.
- Implemented a SimPy core with geospatial plantation ingestion, Delaunay cell generation, battery/charging constraints, drainage-aware routing and Hungarian-assignment task allocation compared with greedy and field-sequential baselines.
- Built a FastAPI/SQLite backend and React/TypeScript dashboard with live WebSocket state, KPI analysis and repeatable parameter sweeps for robot ratios, fleet size and charging-station count.

Autonomous Drone for Soft Payload Deployment and Pickup — **2026**

Four-person university group project | GUI dashboard and communication protocol developer

- Developed the Raspberry Pi 5 telemetry and monitoring stack: MAVLink backend, WebSocket server and browser dashboard for flight state, mission progress, payload status and ArUco-marker visualisation.
- Resolved MAVLink serial contention with a MAVProxy UDP multiplexer, compiled Intel RealSense support for Raspberry Pi from source and tuned telemetry rates to reduce CPU load before full-system integration.
- Co-developed and executed safe autonomous-flight test procedures covering pre-flight readiness, failsafe monitoring, radio-controller/manual overrides, abort sequences and loss-of-control response; integrated testing recorded a 9 N peak payload impact.

OPEN SOURCE & INDEPENDENT ENGINEERING

OrcaSlicer FullSpectrum / Snapmaker Orca Integration **2025 - 2026**

- Contributed mixed-filament colour-mixing and Local-Z dithering with ratdoux and collaborators in OrcaSlicer's C++ codebase. Snapmaker - the 3D-printer manufacturer - merged the co-authored code into its official Snapmaker Orca slicer (PR #409), and upstream OrcaSlicer credits the contribution in mainline PR #13437.
- Applied disciplined Git workflows and collaborative engineering practices across feature integration, testing, documentation and maintenance of a complex cross-platform application.

Custom android updates for Samsung Galaxy Note 5 **2020**

- Built and maintained an official Android 10 ROM for multiple Note 5 variants, working with device trees, kernel and vendor sources, release builds, updates and user support; exceeded 10,000 downloads on the XDA Developers release page.

Armbian Linux Ports for Repurposed Android Phones **2025**

- Ported Armbian to several Android phones and evaluated them as low-cost home servers and Raspberry Pi alternatives, using the built-in battery as a UPS and USB breakout hardware for GPIO/UART connectivity.

Lifestitchr Atlas - Web Deckbuilder **2026**

- Built and deployed a responsive deckbuilder for an original card game, with card search/filtering, deck rules, starter decks, previews, shareable URL state and automated card-atlas generation from source assets.

ADDITIONAL ACADEMIC PROJECTS

- Digital/FPGA design: implemented and tested code converters, multiplexers and demultiplexers in VHDL on an Intel Cyclone IV E FPGA using Quartus.
- Analog design: designed and breadboard-tested BJT amplifier circuits; proficient in LTspice simulation and analysis.
- Embedded: Developed an ESP32 web-based aircraft infotainment prototype.
- Computer Vision: YOLO-based plant-disease identification system.

TECHNICAL SKILLS

Programming: C/C++, Python, TypeScript/JavaScript, VHDL, SQL, HTML/CSS

Embedded & hardware: STM32, ESP32/Arduino, UART, I2C, RS-485/Modbus, Iridium SBD, low-power firmware, sensor integration, FPGA

Software & systems: Git/GitHub, Linux (Archlinux as primary desktop OS), Docker, FastAPI, React, SimPy, SQLite, REST, WebSockets/SSE, testing and documentation

EDA, CAD & fabrication: LTspice, Intel Quartus, Cadence Virtuoso (basic), SolidWorks, Autodesk Fusion, OrcaSlicer and FDM 3D printing

ADDITIONAL TRAINING

- Completed AWS Cloud Architecting coursework as part of APU's cloud computing module.